

8. HEAT CONDUCTION INTO SEMI-INFINITE SOLID

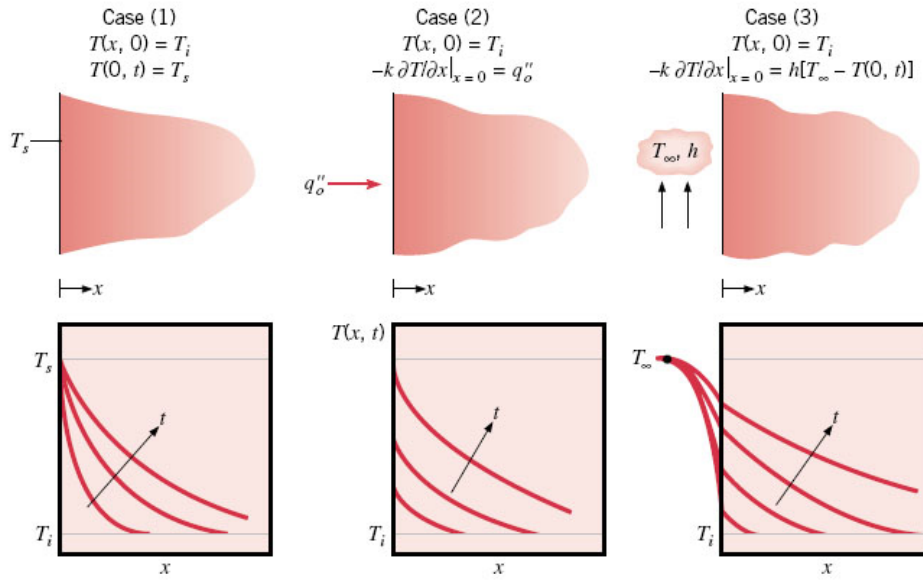


FIGURE 5.7 Transient temperature distributions in a semi-infinite solid for three surface conditions: constant surface temperature, constant surface heat flux, and surface convection.

<CASE I> Constant surface temperature

(1) Similarity solution

$$\boxed{\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2}}$$

$$T(x, 0) = T_i \quad T(0, t) = T_s \quad T(\infty, t) = T_i$$

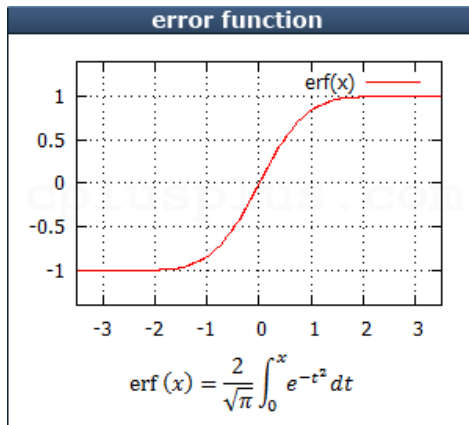
- Similarity variable $\eta \equiv \frac{x}{2\sqrt{\alpha t}}$
- PDE of x and $t \gg$ ODE of η (Using the similarity variable, the PDE has been transformed into an ODE!)

- $\frac{\partial T}{\partial x} = \frac{dT}{d\eta} \frac{\partial \eta}{\partial x} = \frac{1}{\sqrt{4\alpha t}} \frac{dT}{d\eta}$
- $\frac{\partial^2 T}{\partial x^2} = \frac{d}{d\eta} \left[\frac{\partial T}{\partial x} \right] \frac{\partial \eta}{\partial x} = \frac{1}{4\alpha t} \frac{d^2 T}{d\eta^2}$
- $\frac{\partial T}{\partial t} = \frac{dT}{d\eta} \frac{\partial \eta}{\partial t} = -\frac{x}{2t\sqrt{4\alpha t}} \frac{dT}{d\eta}$

$$\frac{d^2 T}{d\eta^2} + 2\eta \frac{dT}{d\eta} = 0 \quad \therefore \quad \boxed{\frac{T - T_s}{T_i - T_s} = \text{erf}\left(\frac{x}{2\sqrt{\alpha t}}\right)} \quad \text{where} \quad \text{erf}(\eta) \equiv \frac{2}{\sqrt{\pi}} \int_0^\eta \exp(-u^2) du$$

$$q_s'' = -k \frac{\partial T}{\partial x} \Big|_{x=0} = -k(T_i - T_s) \frac{d(\text{erf}\eta)}{d\eta} \frac{\partial \eta}{\partial x} \Big|_{\eta=0}$$

$$q_s'' = k(T_s - T_i) \frac{2}{\sqrt{\pi}} \exp(-\eta^2) \frac{1}{2\sqrt{\alpha t}} \Big|_{\eta=0} = \frac{k(T_s - T_i)}{\sqrt{\pi \alpha t}}$$



x	erf(x)	erfc(x)
0	0.000	1.000
0.1	0.112	0.888
0.2	0.223	0.777
0.3	0.329	0.671
0.4	0.428	0.572
0.5	0.520	0.480
0.6	0.604	0.396
0.7	0.678	0.322
0.8	0.742	0.258
0.9	0.797	0.203
1	0.843	0.157
1.1	0.880	0.120
1.2	0.910	0.090

x	erf(x)	erfc(x)
1.3	0.934	0.066
1.4	0.952	0.048
1.5	0.966	0.034
1.6	0.976	0.024
1.7	0.984	0.016
1.8	0.989	0.011
1.9	0.993	0.007
2	0.995	0.005
2.1	0.997	0.003
2.2	0.998	0.002
2.3	0.999	0.001
2.4	0.999	0.001
2.5	1.000	0.000

The above solution shows that $\frac{T - T_s}{T_i - T_s} = 0.843$ at $\eta=1$ ($x = 2\sqrt{\alpha t}$) and 0.995 at $\eta=2$ ($x = 4\sqrt{\alpha t}$)

$\Rightarrow$ If $x \sim \sqrt{\alpha t}$, $T \sim T_i$

$\Rightarrow$ Thermal penetration depth: $\delta_t \sim \sqrt{\alpha t}$

($\delta_t = 2\sqrt{\alpha t}$ with 16% uncertainty, $\delta_t = 4\sqrt{\alpha t}$ with 0.5% uncertainty)

(2) Thermal penetration depth

- Thermal penetration depth (heat diffusion length, heat affected zone: length scale over which temperature change is significant)
- Definition is arbitrary: $\frac{T(\delta_t) - T_s}{T_i - T_s} = 0.99, 0.9, \dots$

$$\frac{T - T_s}{T_i - T_s} = \text{erf}\left(\frac{x}{2\sqrt{\alpha t}}\right)$$

$$\boxed{\delta_t \sim \sqrt{\alpha t}}$$

(3) Alternative approaches to semi-infinite wall problem (**Scaling analysis**)

- Scaling of governing equation: approximate analysis for order of magnitude estimation

$$\frac{\partial T}{\partial t} = \alpha \frac{\partial^2 T}{\partial x^2} \quad \frac{\Delta T}{t-0} \sim \alpha \frac{\Delta T}{\delta_t^2} \quad \boxed{\delta_t \sim \sqrt{\alpha t}}$$

Note $\frac{\partial^2 T}{\partial x^2} = \frac{\partial}{\partial x} \left(\frac{\partial T}{\partial x} \right) \sim \frac{\Delta \left(\frac{\partial T}{\partial x} \right)}{\Delta x} \sim \frac{\left. \frac{\partial T}{\partial x} \right|_{\delta_t} - \left. \frac{\partial T}{\partial x} \right|_0}{\delta_t - 0} \sim \frac{0 - \left(\frac{\Delta T}{\delta_t} \right)}{\delta_t} \sim -\frac{\Delta T}{\delta_t^2}$

(4) Analogy with Stoke's problem of momentum diffusion

$$\boxed{\frac{\partial u}{\partial t} = \nu \frac{\partial^2 u}{\partial x^2}}$$

$$u(x,0) = 0 \quad u(0,t) = U \quad u(\infty,t) = 0$$

Momentum diffusion length scale: $\boxed{\delta_m \sim \sqrt{\nu t}}$

<CASE II> Constant surface heat flux

$$\boxed{T(x,t) - T_i = \frac{2q_0''}{k} \sqrt{\frac{\alpha t}{\pi}} \exp\left(-\frac{x^2}{4\alpha t}\right) - \frac{q_0'' x}{k} \operatorname{erfc}\left(\frac{x}{2\sqrt{\alpha t}}\right)}$$

(Example)

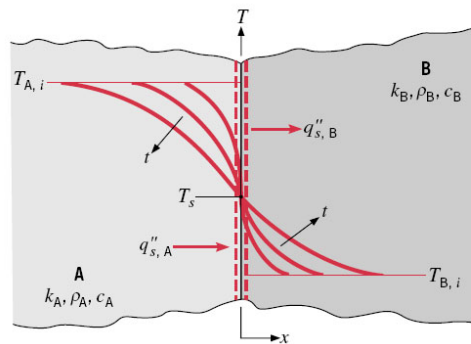


FIGURE 5.9
Interfacial contact between two semi-infinite solids at different initial temperatures.

$$q''_{sA} = q''_{sB} \Rightarrow -\frac{k_A(T_s - T_{Ai})}{\sqrt{\pi\alpha_A t}} = \frac{k_B(T_s - T_{Bi})}{\sqrt{\pi\alpha_B t}}$$

$$T_s = \frac{(\sqrt{k\rho c})_A T_{Ai} + (\sqrt{k\rho c})_B T_{Bi}}{(\sqrt{k\rho c})_A + (\sqrt{k\rho c})_B}$$

Note that $\sqrt{k\rho c}$ is a weighting factor for averaging (thermal effusivity)